

This online appendix supplements the paper "Optimal Power Procurement for Carbon-Free Data Centers Considering Discrete Computing Resource Allocation," submitted to IEEE Transactions on Smart Grid.

APPENDIX A

ELECTRICITY MARKET DATA FOR REPRESENTATIVE DAYS

This appendix presents the hourly spot prices and carbon-free factors for the four seasonal representative days derived from 2025 PJM market data, as described in Section V-A.

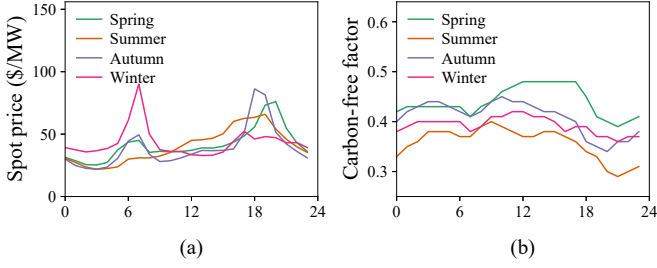


Fig. 1: Hourly spot prices and carbon-free factors for the seasonal representative days. (a) Spot price. (b) Carbon-free factor.

APPENDIX B

CONTRACT DELIVERY PROFILES

This appendix shows the normalized hourly delivery profiles θ_t for the four contract shaping conventions adopted in Section V-B.

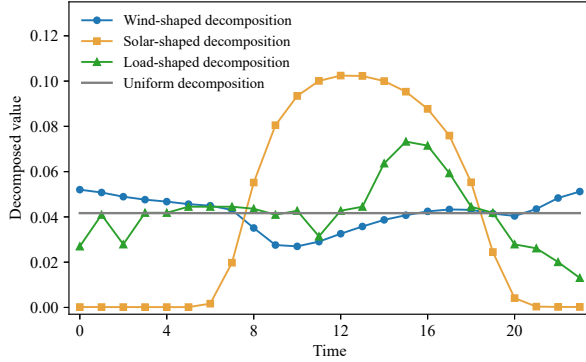


Fig. 2: Normalized hourly delivery profiles of the four contract shaping conventions.